Jason Bak

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EDUCATION

Cornell University: B.S. Mechanical Engineering Ithaca, NY
Cornell Traditions Fellow, GPA: 3.6/4.0 Expected May 2027
Relevant Coursework: Fluid Mechanics, System Dynamics, Mechanics of Engineering Materials, Thermodynamics
Relevant Activities: Cornell University Physical Intelligence, Cornell Cinema Advisory Board, Undergraduate Research

ENGINEERING EXPERIENCE

U.S. Department of Health & Human Services Rhinelander, WI
Facilities Engineering Intern May 2026 - August 2026

- Evaluated and interviewed prospective contractors for a large-scale Tribal health clinic, assessing technical qualifications, project approach, and experience in coordination with Tribal stakeholders.
- Conducted static pressure and fan performance testing on 2 AHUs serving clinical spaces, identifying 10+ deficiencies that were resolved before commissioning sign-off.
- Drafted emergency response plans and egress routes for 6+ health clinics in compliance with NFPA 101, coordinating with facility staff to ensure ADA-compliant paths of egress for patients with mobility constraints.

Cornell University Physical Intelligence Ithaca, NY
Mechanical Team Member August 2025 – Present

- Designed and optimized NACA-series airfoil geometries using SolidWorks and XFOIL, achieving a 14% improvement in thrust-to-weight ratio while meeting competition mass and envelope constraints.
- Performed CFD simulations in ANSYS Fluent to evaluate drag characteristics across 12+ airfoil candidates, reducing parasitic drag by 18% through iterative geometry refinement.
- Developed parametric CAD models for propulsion mounting hardware and structural load-bearing frames, maintaining tolerances within ±0.005" and enabling single-pass assembly integration.

U.S. Department of Health & Human Services Parker, AZ
Civil/Fluids Engineering Intern May 2025 - August 2025

- Designed and modeled 15+ site infrastructure systems in AutoCAD and Civil 3D, including pressurized water distribution networks, utility layouts, and structural rehabilitation plans, delivering engineering packages that secured $500k in federal funding
- Conducted field inspections and diagnostics across 20+ sites, pressure logging, flow testing, structural assessment — to characterize system performance and inform 8 corrective design interventions
- Performed structural load and stress analysis in ANSYS to validate rehabilitation designs for a water storage tower serving 300+ residents, reducing projected water loss by over 70%

PROJECTS

Havasupai Water Tower Rehabilitation Havasupai, AZ

- Diagnosed a critical structural failure in a 50,000-gallon elevated water storage tank through on-site inspection and finite element analysis, identifying stress concentrations at weld joints exceeding allowable limits by 35%.
- Modeled and validated structural rehabilitation designs in ANSYS and AutoCAD, reducing maximum von Mises stress by 42% and restoring the tank to a projected 25-year service life.

CUStudy Application Ithaca, NY

- Engineered an IoT occupancy monitoring system using Arduino and C to detect Wi-Fi-connected devices across 3 Cornell library locations, then trained a model in Python that predicted real-time student occupancy with 91% accuracy against 2,000+ manual headcount samples.
- Designed custom sensor enclosures in SolidWorks with optimized antenna placement to maximize a 25-meter detection range, and built a live Flask dashboard serving 150+ daily active users during finals week.

TECHNICAL SKILLS

CAD & DESIGN
AutoCAD · SolidWorks · ANSYS(FEA/CFD) · OpenRocket

PROGRAMMING & ANALYSIS
MATLAB · Python · SQL · Java

DOCUMENTATION & TOOLS
3D Printing · Hand Tools · GD&T · Engineering Drawings

PROFESSIONAL SKILLS
Technical Communication · Problem Solving · Public Speaking